

Week 01 Report

Cybersecurity
Internship

2026

WEEK 01: Security Assessment Report – User Management Web Application

1. Application Setup

During **Week 1**, I set up a Node.js-based User Management web application downloaded from GitHub for cybersecurity testing. The setup steps included:

- Extracting the project files and navigating to the project directory
- Installing dependencies using **npm install**
- Running the application using `node app.js`
- Accessing the application at <http://localhost:3000>
- Exploring key pages such as Signup, Login, and Dashboard

This setup provided a working environment to perform vulnerability assessment and security testing.

2. Tools Used

- OWASP ZAP for automated vulnerability scanning
- Browser Developer Tools for manual testing
- Manual input-based testing techniques

3. Testing Methodology

- **XSS Testing**
Payload used: `<script>alert('XSS')</script>`. No execution occurred.
The application is not vulnerable to basic XSS.
- **SQL Injection**
Payload used: `admin' OR '1'='1`. Login failed.
- **Input Validation**
Invalid inputs were rejected. Proper validation is implemented.
- **Password Policy**
Weak passwords were rejected. Password rules are enforced.
- **Brute Force Testing**
Multiple incorrect attempts were allowed without restriction.

4. **Vulnerabilities Identified**

- The application is **not vulnerable to basic XSS attacks**
- SQL Injection attempts were **unsuccessful**
- Input validation is **properly implemented**
- Password policy is **enforced correctly**
- The application is vulnerable to brute-force attacks **due to unlimited login attempts**

5. **Secure Features**

Input validation, password policy, and protection against XSS and SQL injection are properly implemented.

Conclusion

The application has basic security controls but lacks protection against brute-force attacks and has multiple misconfigurations. Improvements are recommended. This assessment helped identify both secure features and areas where the application requires improvement.